

Mutual Funds Analytics

Individual Capstone Project |
Bluestock Fintech Pvt. Ltd.
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July 2026

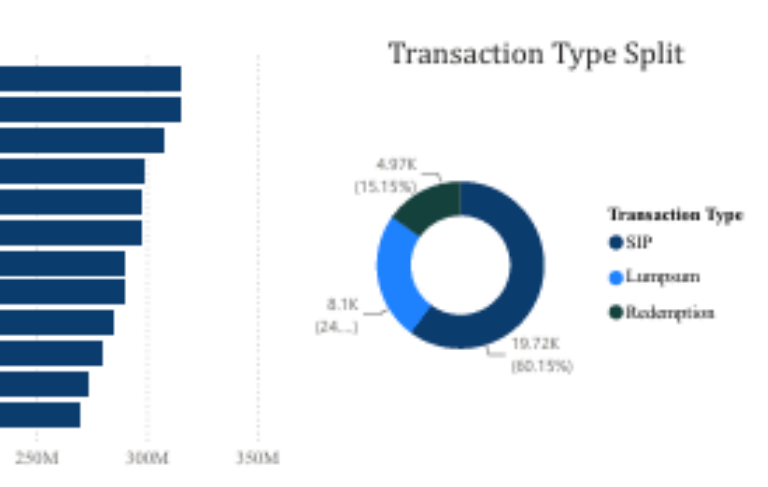
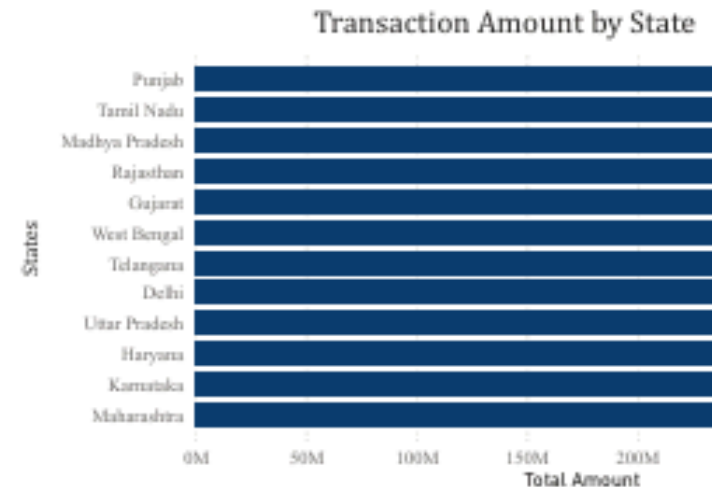
Fund Performance

Scheme Name	AMC	3 Year Return	Sharpe	Alpha	Max Drawdown	Expense Ratio	Fund Score
Axis Bluechip Fund - Regular - Growth	Axis Mutual Fund	11.84	0.85	1.41	-0.14	1.64	100.00
HDFC Top 100 Fund - Regular Plan - Growth	HDFC Mutual Fund	14.84	1.06	0.78	-0.25	1.55	89.44
Kotak Bluechip Fund - Regular - Growth	Kotak Mahindra MF	12.25	0.87	1.27	-0.18	1.59	85.94
UTI Mid Cap Fund - Regular - Growth	UTI Mutual Fund	15.61	0.82	1.12	-0.28	1.51	85.70
ABSL Liquid Fund - Regular - Growth	Aditya Birla Sun Life MF	5.14	5.14	1.18	0.00	0.79	85.62
SBI Magnum Gilt Fund - Regular Plan - Growth	SBI Mutual Fund	6.07	1.52	1.60	-0.04	0.77	83.56
Axis Bluechip Fund - Direct - Growth	Axis Mutual Fund	12.14	0.87	1.43	-0.22	0.75	83.40
Nippon India Gilt Securities Fund - Regular - Growth	Nippon India MF	5.31	1.33	0.89	-0.08	0.55	82.45
HDFC Short Term Debt Fund - Regular - Growth	HDFC Mutual Fund	7.37	1.84	1.98	-0.04	0.56	81.49
DSP Top 100 Equity Fund - Regular - Growth	DSP Mutual Fund	12.82	0.92	1.82	-0.15	1.54	81.10
Kotak Emerging Equity Fund - Regular - Growth	Kotak Mahindra MF	18.23	0.96	1.91	-0.24	1.56	77.68
ABSL Small Cap Fund - Regular - Growth	Aditya Birla Sun Life MF	22.38	0.90	1.84	-0.35	1.53	76.17
Axis Small Cap Fund - Regular - Growth	Axis Mutual Fund	20.98	0.84	0.51	-0.52	1.38	75.30

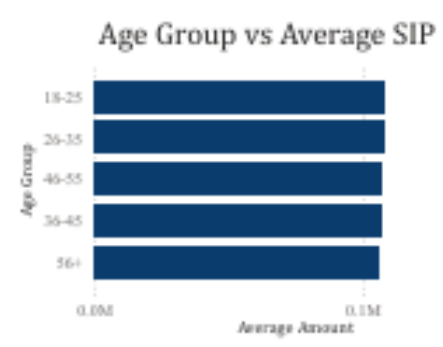


- AMC
- ☐ Aditya Birla Sun Life MF
 - ☐ Axis Mutual Fund
 - ☐ DSP Mutual Fund
 - ☐ HDFC Mutual Fund
 - ☐ ICICI Prudential MF
 - ☐ Kotak Mahindra MF
 - ☐ Mirae Asset MF
 - ☐ Nippon India MF
 - ☐ SBI Mutual Fund
 - ☐ UTI Mutual Fund
- Category
- ☐ Debt
 - ☐ Equity
- Plan
- ☐ Direct
 - ☐ Regular
- Year, Quarter, Month, Day
- ☐ 2022
 - ☐ 2023
 - ☐ 2024
 - ☐ 2025
 - ☐ 2026

Investor Analytics



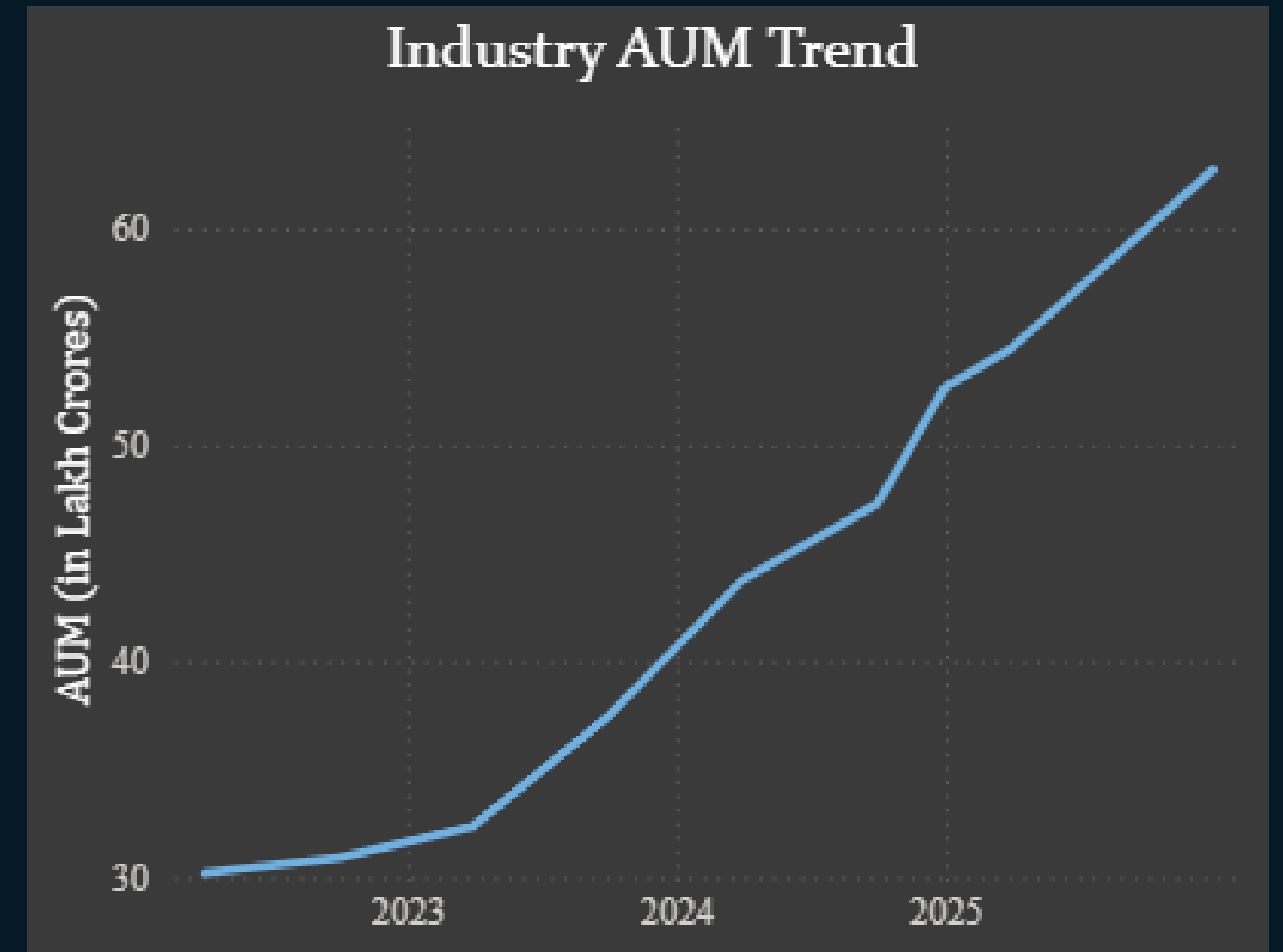
- State
- ☐ Delhi
 - ☐ Gujarat
 - ☐ Haryana
 - ☐ Karnataka
 - ☐ Madhya Pradesh
 - ☐ Maharashtra
 - ☐ Punjab
 - ☐ Rajasthan
 - ☐ Tamil Nadu
 - ☐ Telangana
 - ☐ Uttar Pradesh
 - ☐ West Bengal
- Age Group
- ☐ 18-25
 - ☐ 26-35
 - ☐ 36-45
 - ☐ 46-55
 - ☐ 56+
- City Tier
- ☐ B30
 - ☐ T30



Executive Summary & Objective

Executive Summary

- The Industry Milestone: By December 2025, the Indian mutual fund landscape scaled to Rs. 81 lakh crore in total Assets Under Management (AUM), with monthly SIP inflows achieving an all-time high of Rs. 31,002 crore across 26.12 crore active folios.
- Project Mission: To solve data fragmentation and accessibility issues for individual retail investors and corporate wealth managers by developing an automated pipeline and visual intelligence dashboard.
- Key Engineering Goals: Build an automated ETL ingestion mechanism, construct a production-ready relational schema, run statistical multi-period calculations, and implement interactive user drill-downs.



Problem vs. Engineering Solution

Observed Industry Friction

Data Fragmentation: AMFI datasets are trapped across disparate daily TXT logs, quarterly PDFs, and raw web elements.

Performance Context Gap: Simple daily NAV tracking limits an investor's ability to evaluate performance on a risk-adjusted scale.

Benchmark Tracking Void: Advisors struggle to cleanly map index tracking errors or recognize true alpha anomalies.

Investor Behavior Blind Spot: Asset management companies lack deep visibility into regional distribution trends.

Delivered Architectural Fix

Developed a consolidated Python-based engine linking flat files directly with public REST APIs.

Engineered custom analytical models calculating rolling annualized returns alongside traditional market parameters.

Integrated active equity fund movements alongside the daily closing prices of Nifty 50 and Nifty 100 indices.

Segmented anonymized transaction registers by demographic age brackets, city designations, and state boundaries.

Data Engineering Layer & Star Schema

The Data Foundation

Aggregated, processed, and validated over 87,000+ active transaction logs and 4.5 years of underlying NAV tracking points (ranging from January 2022 to May 2026).

Star Schema Database Design

- Load operations feed a highly performant 5-table schema designed to reduce computing overhead during dashboard operations:
- `dim_fund`: Scheme classification properties, AMC brand fields, expense boundaries, and management names.
- `dim_date`: Time intelligence mappings across years, quarters, months, and operational business days.
- `fact_nav`: Daily price variations tracking core capital movements and immediate percentage shifts.
- `fact_transactions`: Granular ledger of user behaviors capturing volume traits across geographic nodes.
- `fact_performance`: Consolidated index anchoring all calculated risk variables.

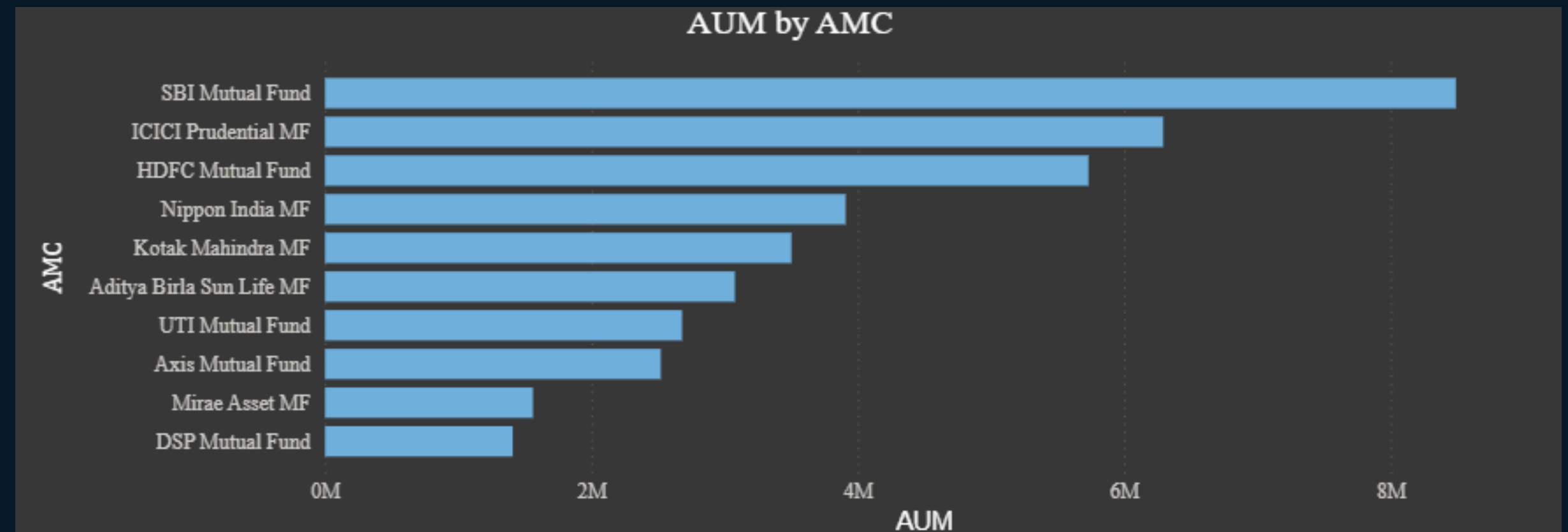
Data Refinement

Handled trading anomalies using forward-fill data cleaning techniques to align trading schedules across dates.

Macro Industry Insights

Macro Industry Insights

- Core High-Level Metrics:
- Total System AUM: Recorded at 391.76 across tracked channels.
- System Folios: Established a total customer base of 416.42 million accounts.
- SIP Monthly Momentum: System-wide inflows calculated at 940K active transactional iterations.



Asset Management Dominance:

Cleanly verified SBI Mutual Fund's commanding lead in asset share, followed in order by ICICI Prudential Mutual Fund and HDFC Mutual Fund.

Composite Index Highlights

- Axis Bluechip Fund (Regular - Growth): Led the composite scorecard rankings with a perfect score of 100.00, yielding an 11.84% 3-year return profile balanced against an alpha of 1.41 and a contained maximum drawdown of -0.14.
- HDFC Top 100 Fund (Regular - Growth): Achieved a score of 89.44, displaying a strong 3-year return of 14.84% and a Sharpe ratio of 1.06.
- Kotak Bluechip Fund (Regular - Growth): Positioned firmly with a score of 85.94, supported by an alpha generation rate of 1.27.

Asset Management Dominance:

Aggressive equity variants such as the ABSL Small Cap Fund reached higher multi-year absolute returns 22.38%, but faced steeper max drawdown limits (-0.35), shifting their positioning inside risk-adjusted scoring logic.

Fund Performance Profiles

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Axis Small Cap Fund - Regular - Growth	Axis Mutual Fund	20.98	0.84	0.51	-0.52	1.38	75.30
ICICI Pru Bluechip Fund - Regular - Growth	ICICI Prudential MF	11.54	0.82	0.66	-0.14	1.42	73.55
UTI Flexi Cap Fund - Regular - Growth	UTI Mutual Fund	15.34	0.96	1.79	-0.22	1.64	73.47
Nippon India Small Cap Fund - Regular - Growth	Nippon India MF	20.15	0.81	0.80	-0.23	1.53	72.68
Kotak Liquid Fund - Regular - Growth	Kotak Mahindra MF	6.18	6.18	1.52	0.00	0.60	72.60
HDFC Mid-Cap Opportunities Fund - Direct - Growth	HDFC Mutual Fund	15.29	0.80	0.90	-0.21	0.78	70.37
ICICI Pru Value Discovery Fund - Regular - Growth	ICICI Prudential MF	14.76	0.98	0.55	-0.20	1.41	67.20
SBI Small Cap Fund - Direct Plan - Growth	SBI Mutual Fund	23.14	0.93	1.13	-0.53	0.72	66.80
UTI Nifty 50 Index Fund - Regular - Growth	UTI Mutual Fund	12.10	0.93	0.93	-0.11	1.57	66.72
Nippon India ETF Nifty 50 BeES	Nippon India MF	11.77	0.91	1.80	-0.12	0.89	65.77
Nippon India Large Cap Fund - Direct - Growth	Nippon India MF	12.33	0.88	1.70	-0.19	0.72	64.73
HDFC Top 100 Fund - Direct Plan - Growth	HDFC Mutual Fund	13.38	0.96	1.13	-0.15	0.92	62.91
ICICI Pru Liquid Fund - Regular - Growth	ICICI Prudential MF	7.68	7.68	1.85	0.00	0.74	62.91

Ingestion Computations

Multi-Period Historical Benchmarking

- Scheme 100033: Achieved an impressive 19.92% CAGR over 1,607 structured market trading sessions, backed by an alpha calculation of 0.2719 against index benchmarks.
- Scheme 101206: Delivered an absolute return profile of 15.70% CAGR, generating a reliable alpha value of 0.2139.

Market Sensitivity Frameworks

Low beta indicators recorded for Scheme 100016 (-0.058) and Scheme 101207 (-0.065) validate effective diversification mechanics, moving independently of macro market swings.

	amfi_code	Start_NAV	End_NAV	Trading_Days	CAGR
0	100016	520.4608	583.6113	1607	0.018121
1	100025	26.3169	31.8843	1607	0.030550
2	100033	107.3758	342.0072	1607	0.199217
3	101206	305.0996	773.2939	1607	0.157011
4	101207	38.5736	53.9836	1607	0.054121

	amfi_code	Sharpe
34	148567	1.068224
30	120843	0.965561
36	148569	0.919047
25	120505	0.883256
19	119551	0.860977
38	149323	0.832885
2	100033	0.808268
9	118632	0.758851
16	119094	0.730547
3	101206	0.717409

Advanced Risk Insulation: Value at Risk

Downside Probability Forecasts

Evaluated absolute tail-end structural capital hazards using a 95% statistical testing threshold:

- Scheme 101207: Logged an explicit 95% Historical VaR limit of -2.39%, indicating a minor probability of daily capital adjustments expanding past that threshold under typical trading conditions.
- Scheme 119095: Revealed a tightly matched 95% daily value fluctuation floor calculated at -2.32%.

Conditional Value at Risk (CVaR) Modeling

When tracking anomalous events past standard confidence bands, the engine's CVaR layer captures average extreme variations extending down to -3.02% (Scheme 101207) and -3.01% (Scheme 119599).

	amfi_code	VaR_95	CVaR_95
4	101207	-0.023915	-0.030289
17	119095	-0.023284	-0.029690
22	119599	-0.023155	-0.030163
11	118634	-0.022810	-0.029940
39	149324	-0.021520	-0.028573
21	119598	-0.021502	-0.028444
16	119094	-0.016997	-0.022375
29	120842	-0.016950	-0.021251
2	100033	-0.016902	-0.021850
7	102886	-0.016857	-0.021771

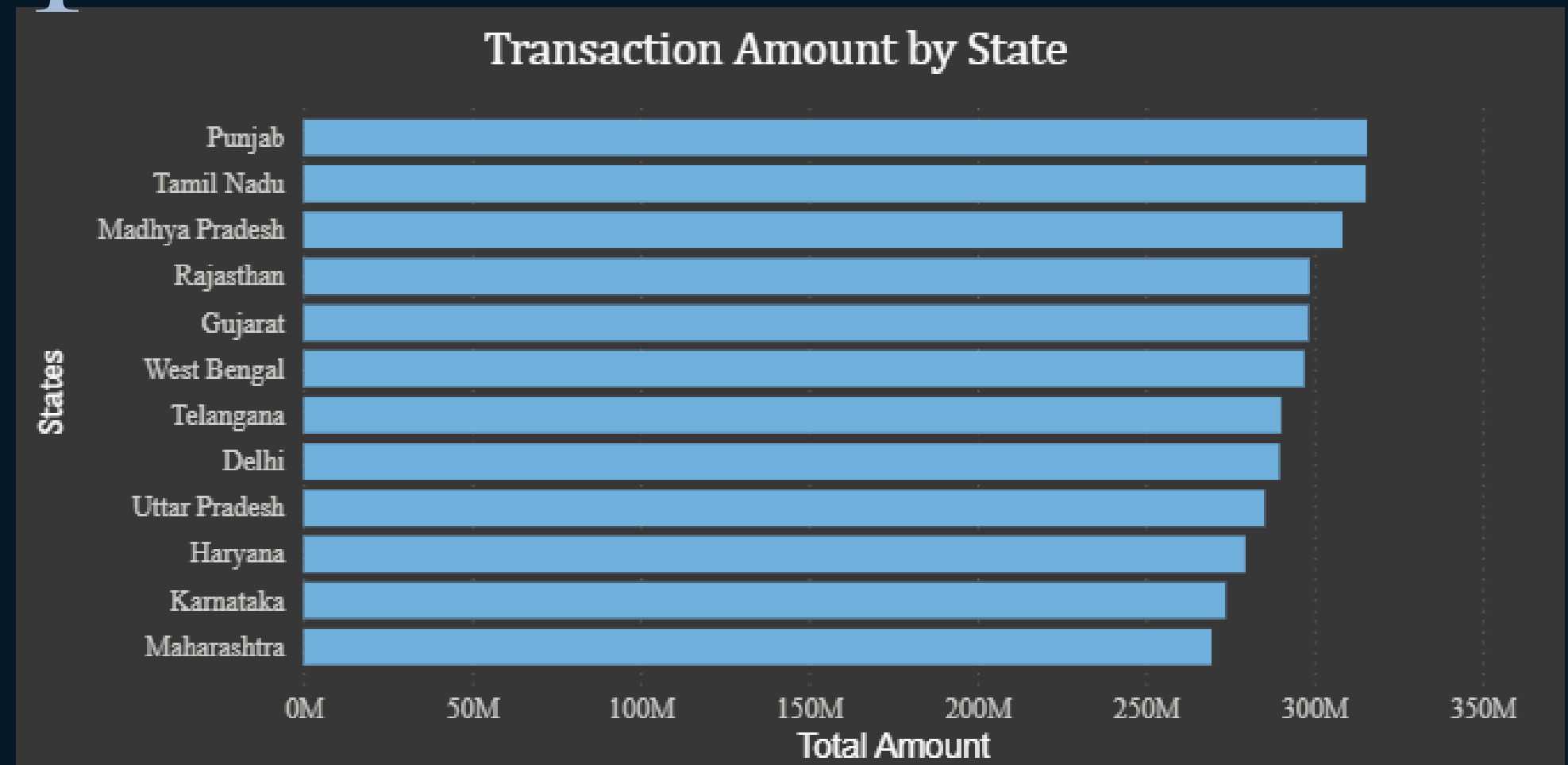
Investor Demographics

Geographic Wealth Distribution

Capital allocation concentration metrics highlight high-volume performance out of Punjab, closely supported by strong retail participation rates across Tamil Nadu and Madhya Pradesh.

Transactional Method Splitting

- SIP Dominance: Systematic Investment Plans act as the main driver for long-term customer acquisition, accounting for an impressive 60.15% share of total platform transaction structures (19.72K} active ledgers.
- Lumpsum Allocations: Account for a solid 24.70% structural share, tracking larger, opportunistic market entries.



Demographic Alignment

Cross-referencing customer demographics confirms stable regular SIP averages across all adult age groups, with younger investors showing steady monthly commitment rates.



Product Trends & Asset Allocation

Product Flow Dynamics

- Liquid Assets Category: Commands an outsized volume share inside the architecture, directing a massive \$451,275\$ inside regional transaction flows.
- Flexi Cap Strategy Implementations: Represent the primary choice for diversified equity investing, bringing in a total allocation flow of \$63,989\$.
- Mid Cap & Large/Mid Cap Funds: Maintain strong secondary placement positions, tracking total flow inputs of \$55,312\$ and \$57,752\$ respectively.

Strategic Takeaway

While capital preservation strategies draw the largest total volume, flexible and thematic equity structures remain the primary choice for investors seeking outsized market alpha.

Conclusions & Project Extensions

Replaced outdated static financial tracking methods with an agile, end-to-end automated platform that translates raw historical NAV points into actionable risk-adjusted investment signals.

On Process: Automated HTML email report generator sending weekly performance summaries

